Triple Offset Butterfly Valves

TOV Series

Pressure Range:
150/300 Class

Size Range:
$2^{1/2}$"–48"

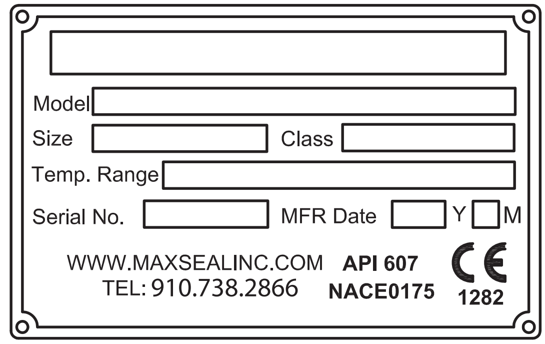
Max-Seal, Inc.
4815 West 5th St.
Lumberton, NC 28358

P.O. Box 1293
Lumberton, NC 28359
www.maxsealinc.com

910-738-2866
910-739-1733
sales@maxsealinc.com

Construction Design Specifications

Product Identification



Triple Offset Valves

Service

Max-Seal high quality butterfly valves are suitable for both on/off and throttling service.
The triple offset design incorporates torque seat for tight shutoff or control even in the roughest applications.
Industries where the triple offset butterfly valves are widely used include Power, Refining, Petrochemical, Chemical, and Pulp and Paper and many others.

FEATURES

Shaft Bearings

Shaft is centered on two self-lubricating bearings with extremely high rotation precision and stability. Bearing is PTFE lined on internal surface on bronze sintered stainless steel base material with super abrasion resistance and self-lubrication to prevent possible seizure problems.

Compact Construction

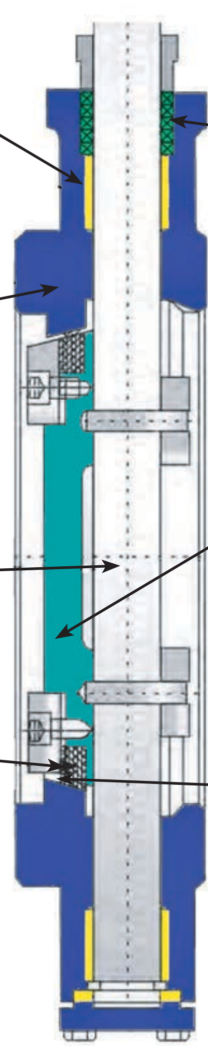
Due to the nature of their design, butterfly valves can offer very short face-to-face dimensions. Less metal means butterflies are often a very economical alternative to ball valves. Also, reduced torque for butterfly valves offers even more cost savings, as they can be operated with smaller actuators.

Low Emission Shaft

Shafts are precisely ground by CNC machine with very fine finish and low friction factor, therefore, low emission from packing and longer packing cycle life.

Various Sealing Design

Valves have standard laminated seal ring with graphite layers carefully assembled between stainless steel layers with reliable high temperature resistance and seat tightness. Torque seating during closing of the valve provides friction free sealing for long cycle life. Based on suitability of the intended service, seat options may include PTFE seat or metal seat. Consult Max-Seal with specifics on your application, and we'll be happy to help select a valve for your service.



Stem Seal

Stem seal assembly is live loaded with two Belleville Springs. This ensures continuous compression of packing and sealing. Rocker shaped gland bridge compensated for uneven adjustment of gland bolts. Adjustable stem packing with multiple graphite rings seal on high

Disc

Disc is designed with a profile to minimize resistance to flow and pressure drop across the valve and maximize the flow capacity.

Seat

Seat is integral on body and is hard faced with Stellite of suitable alloy. Seat is precision machined to ensure perfect match with the seal ring. This provides bubble tight seal, prevents galling and friction during seating and unseating, provides resistance to erosion during high velocity fluid flow and prevents corrosion due to media.

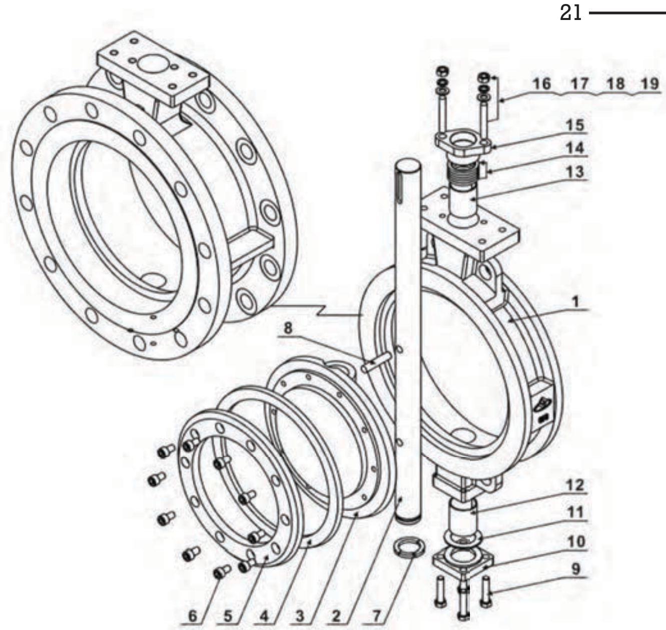
Triple Offset Eccentric Valve Model Number ID Codes

Model	Pressure Class	Body		Disc		Stem		Seat		Seal Ring		Operator		Size in/mm
Triple Wafer-BWTL Flanged-BFTL Lug-BLTL	150	316 SL CF3M	SL	316 SL CF3M	SL	XM-19	XM	CF3M	SL	316SS + Graphite	SG	Gear	G	3"/80
	300	WCB	CS	WCB	CS	410SS	S1	13% Chrome	CR	Inconel 625 + Graphite	IG	Bare Stem	N	4"/100
	600	WC6	W6	WC6	W6	XM-19	XM			316SS Solid Metal	SM	Actuator	A	6"/150
	900													8"/200

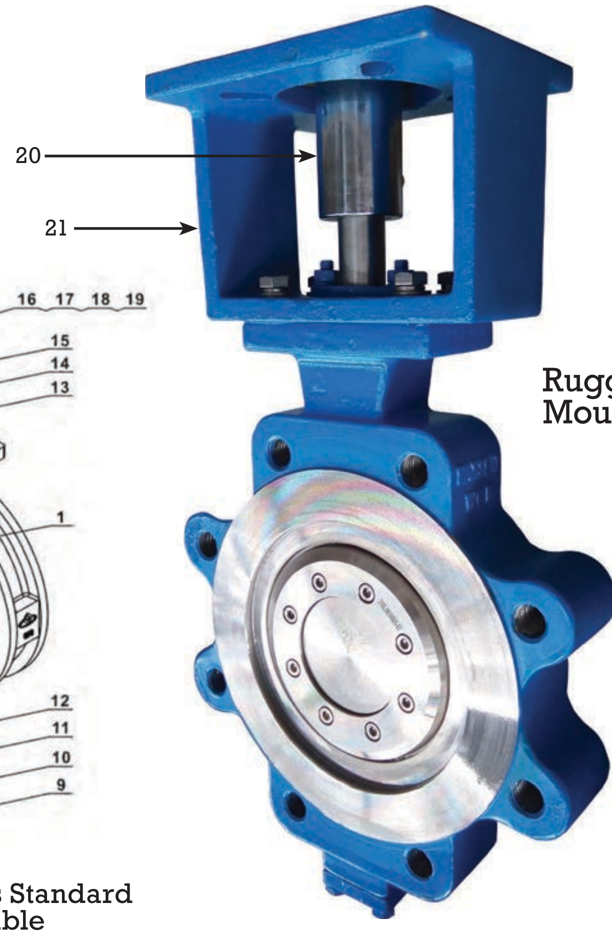
(Always add "T" for triple offset)

Standard Bill of Materials

Exploded View
Parts List



Optional: Intregal Body Seat/ 316SS As Standard
Optional Seats Available



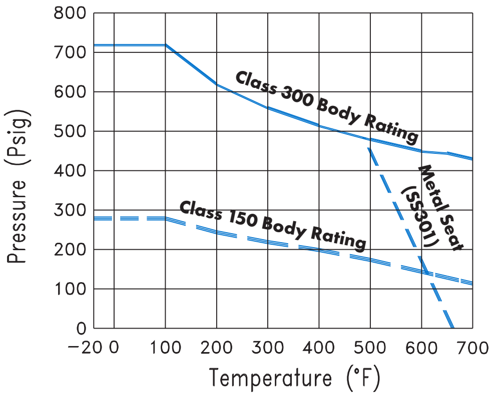
Rugged Heavy Duty
Mounting Design

Parts List / Materials of Construction

No	Parts	Qty	Standard Material	Optional Material
1	Body	1	**316SL, WCB**	CF8, CF8M
2	Shaft	1	**XM-19**	17-4PH, SS304, SS316, SS410
3	Disc	1	**CF3M**	WCB, CF8, CF8M
4	Sealing Ring	1	**316SL + Graphite**	PTFE
5	Retainer Flange	1	**CF3M**	Carbon Steel, SS304, SS316
6	Hexagon Bolt	as required	**SS304**	SS316
7	Split Collar	1	**SS304**	SS316
8	Pin	2	**SS316L**	SS304, SS316
9	Hexagon Bolt	4	**SS304**	SS316
10	Blind Flange	1	**CF8M, WCB**	CF8, CF8M
11	Gasket	1	**SS304**	Spiral Wound Metal/Graphite Gasket
12	Self-Lubricating Bearing	1	**SS304 + N**	Composite Material
13	Self-Lubricating Bearing	1	**SS304 + N**	Composite Material
14	Packing	as required	**Graphite**	PTFE
15	Gland	1	**CF8M, WCB**	WCB, CF8, CF8M
16	Stud	2	**SS304**	SS304
17	Washer	2	**SS304**	SS304
18	Spring Washer	2	**SS304**	SS304
19	Nut	2	**SS304**	SS304
20	STPM ISO Coupler	1	**SS304**	SS304
21	Actuator Mountin Bracket ISO		**SS304/WCB**	SS304/WCB

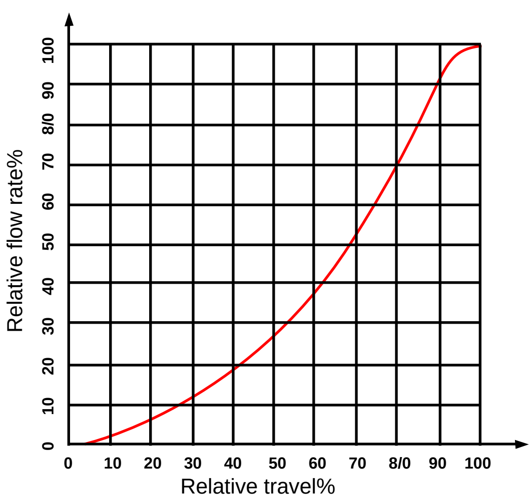
Valve Flow Coeffications and Technical Data

Pressure Temperature Rating



Note: WCB is not recommended for prolonged use above 800° F

Flow Characteristic Curve



Rated Cv

Size mm	Size inch	Rated Cv
50	**2**	95
65	**2½**	175
80	**3**	270
100	**4**	470
125	**5**	850
150	**6**	1400
200	**8**	2300
250	**10**	3560
300	**12**	5800
350	**14**	7800
400	**16**	10000
450	**18**	13500
500	**20**	18000
600	**24**	25000
700	**28**	38500
800	**32**	48000
900	**36**	52500
1000	**40**	61700
1200	**48**	81000

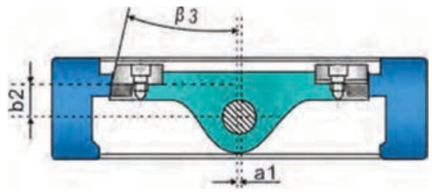
Fugitive Emission Capabilities

In a modern chemical processing plant or refinery, a major chunk (approx. 60%) of VOC (volatile organic compounds) emissions are ascribed to valve leakages, and of these, almost 80% are discharged at the valve stems. The remaining portion of valve failures is linked to leaks from the bonnet, end connections, and seats. Due to day by day changing environmental regulations and strict climate change policies globally, having a valve certified to fugitive emissions make the industrial facilities safe and efficient. Max-Seal Triple Offset Butterfly Valve inherits this capability and is, therefore, the right choice for a broad spectrum of applications associated with VOC releases.

Cv Versus Valve Rotation, in Degrees

Size mm	Size inch	Cv 10°	20°	30°	40°	50°	60°	70°	80°	90°
50	**2**	1.53	4.29	10	12	16	20	42	82	95
65	**2½**	3.32	8.36	20	23	27	44	86	159	175
80	**3**	4.68	14.78	27	38	39	79	152	249	270
100	**4**	9.97	39.44	81	168	231	307	359	425	470
125	**5**	54.06	164.04	270	391	473	531	634	781	850
150	**6**	58.33	172.22	307	501	614	953	1183	1379	1400
200	**8**	66.74	185.38	360	619	983	1409	1793	2171	2300
250	**10**	147.95	385.90	710	1106	1646	2305	2892	3419	3560
300	**12**	254.47	622.68	1078	1636	2419	3470	4435	5332	5800
350	**14**	341.64	820.42	1365	2198	3234	4375	7158	7165	7800
400	**16**	408.16	965.28	1654	2694	4041	5833	7634	9573	10000
450	**18**	489.22	1175.31	2015	3305	4982	7238	9737	12076	13500
500	**20**	692.40	1539.74	2560	4013	6321	9214	12510	15716	18000
600	**24**	1027.97	2279.72	3840	5858	8934	12862	17425	22261	25000
700	**28**	1274.25	2523.27	4628	7812	12254	18887	26357	33522	38500
800	**32**	1811.41	4124.70	6387	10453	15803	23502	33052	42317	48000
900	**36**	2463.56	4123.04	6909	11359	16879	25188	35844	44836	52500
1000	**40**	4225.34	11579.49	13908	16501	21419	30477	37120	47632	61700
1200	**48**	5264.63	17148.26	20640	26114	30179	37181	46032	52688	81000

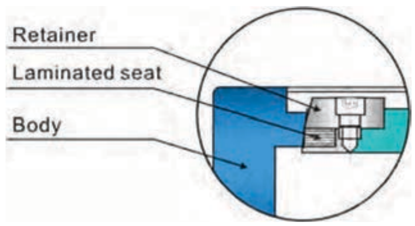
Butterfly Valve | Design Characteristics



Triple Offset Design Principle

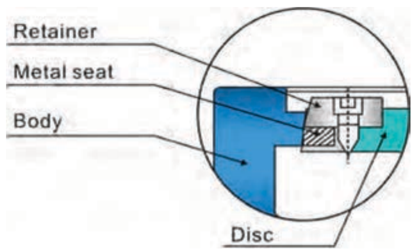
1. Shaft centerline is offset from the centerline of the valve and of the pipe.
2. Shaft is offset behind the disc face.
3. The third offset is at the sealing face. The sealing surface on the disc is similar to a slice taken from a con shape. This cone shape is offset from the centerline of the pipeline. The shape resulting from this third offset reduces friction and thus torque, and helps provide for uniform sealing forces around the surface of the seat.

Features of Various Seat Designs



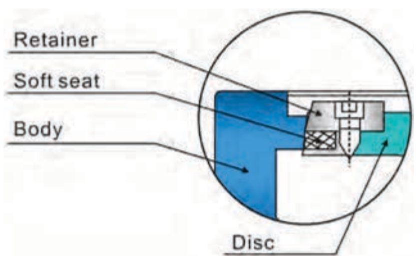
Laminated Seal Ring

1. Low operation torque and easy operation.
2. Tight shut off up to bubble tightness.
3. Reliable tightness can be ensured even under low differential pressure.
4. Good pressure, corrosion, and abrasion resistance and long cycle life.



Metal Seat

1. Low operation torque and easy operation.
2. Hard alloy coated metal seat provides super abrasion resistance and high temperature service.
3. Excellent pressure, corrosion, and abrasion resistance and long cycle life.



Soft Seat

1. Reliable bubble tightness.
2. Low operation torque and easy operation.

Design Specifications

Nominal Diameter:

DN50-DN1200 **2.5" - 48"**

Pressure Rating:

PN10-PN40, **ANSI 150, ANSI 300**

Connection Type:

Wafer, Lug, Flanged

Working Temperature:

-40°C~+160°C / **-40° F~+320° F** for PTFE seat

-40°C~+350°C / **-40° F~+662° F** for metal seat

For temperature exceeds the above ranges, please consult factory.

Design and manufacture standard:

API609

Face-to-face dimensions standard:

API609 for flanged type, EN558.1 for wafer and lug type.

Seat tightness:

Control valve-ANSI/FCI 70.2-2006 Class V

On/Off Valve-ISO 5208 rate D for metal seat, ISO 5208 rate A for soft seat.

For higher tightness requirement, please consult factory.

Certifications:

Firesafe


CERTIFICATE
API STANDARD 607 SEVENTH EDITION, JUNE 2016


CERTIFICATE
API STANDARD 607 SEVENTH EDITION, JUNE 2016

ISO 9001


TÜV NORD

CERTIFICATE

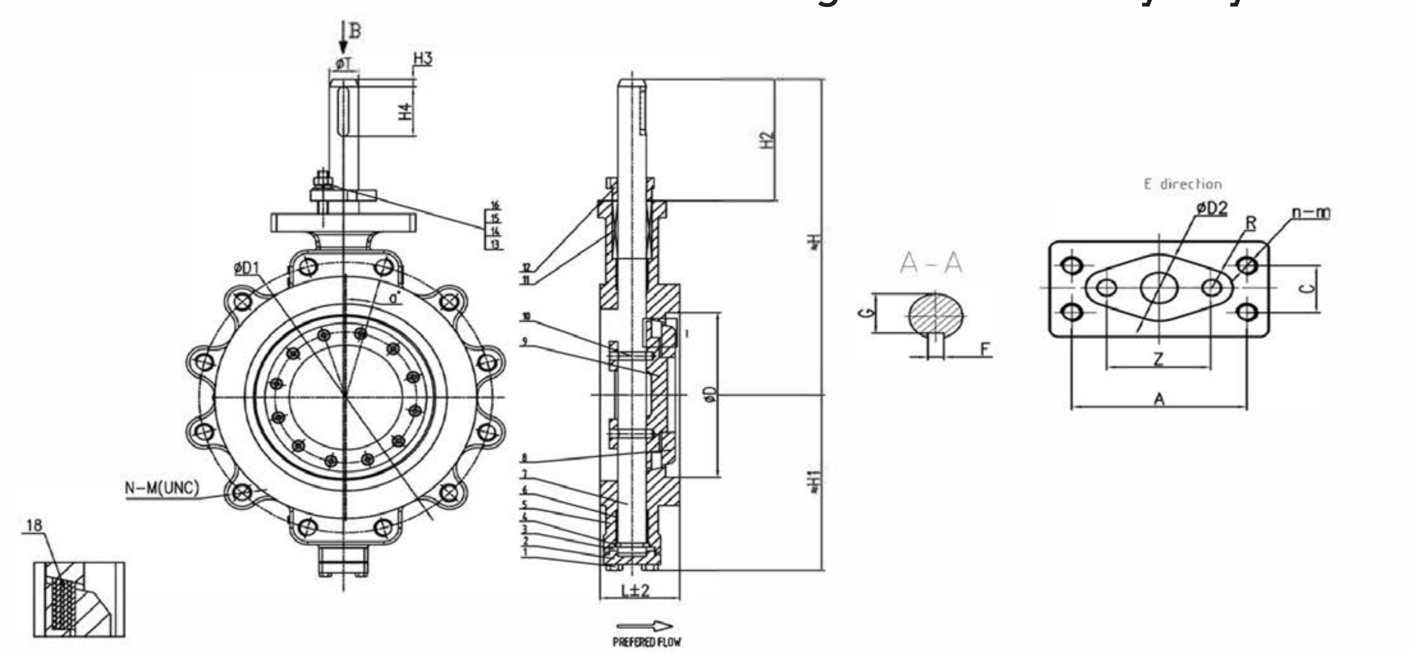
Management system as per
GB/T 19001-2016/ ISO 9001 : 2015

Zhejiang Linuo Flow Control Technology Co., Ltd.

Design and Production of Ball Valves, Butterfly Valves, Control Valves, Gate Valves, Globe Valves and Penumatic Actuators

Butterfly Valve | **Class #150**

Dimensional Data Actuator Mounting With Stem Keyway



Verify Mounting Dimensions Before Manufacturing Mounting Hardware.

Bare Shaft Valves

Size mm/inch	L			Actuator Mounting Dimensions																		Torque (in-lbs)	Weight (lbs)
	Wafer	Lug	Flanged	øD	øD1	N-M	a°	H	H1	H2	H3	H4	øT	G	F	A	Z	n-m	C	øD2	R		
3”	49/1.93	48	CF	84	152.4	4”-5/8”-11”	45°	218	119	77	3	25	16	12	5	80	40	2-M12	30°	28	R8	761	21
4”	56/2.20	54	CF	105	190.5	8”-5/8”-11”	22.5°	232	142	77	3	25	20	16.5	6	90	44	4-M12	30°	32	R9	1416	CF
6”	70/2.76	57	CF	157	241.5	8”-3/4”-10”	25.5°	276	168	93	3	35	25	21	8	90	50	4-M10	40°	38	R10	2436	51
8”	71/2.80	64	CF	205	298.5	8”-3/4”-10”	25.5°	307	209	95	3	40	30	25	10	100	58	4-M10	40°	44	R11	4202	78
10”	76/2.99	71	CF	250	362	12”-7/8”-9”	15°	350	240	110	3	50	35	30	10	130	64	4-M16	50°	52	R12	7035	115
12”	83/3.27	81	CF	300	300	12”-7/8”-9”	15°	380	266	110	4	50	40	35	12	130	68	4-M16	50°	56	R11	12168	164
14”	92/3.62	92	CF	350	476	12”-1“-8”	15°	450	296	150	4	60	45	39.5	14	134	102	4-M16	64°	75	R16	17699	217
16”	102/4.02	102	CF	400	540	16”-1“-8”	11.25°	490	338	150	4	60	50	44	16	134	102	4-M16	64°	75	R16	26549	CF
18”	114/4.49	114	CF	450	578	16”-1 1/8”-8”	11.25°	535	367	151	5	80	55	49	16	175	120	4-M20	70°	90	R16	33186	489
20”	127/5.00	127	CF	500	635	20”-1 1/8”-8”	9°	574	400	151	5	80	55	49	16	175	120	4-M20	70°	90	R16	45133	688
24”	154/6.06	154.69	CF	450	749.5	20”-1 1/4”-8”	9°	638	456	163	5	80	65	58	18	210	145	4-M24	90°	102	R20	33186	CF

2 1/2" SIZE AVAILABLE CONSULT FACTORY FOR DIMENSIONAL INFORMATION.

Gear OPS + 150 Class Valve

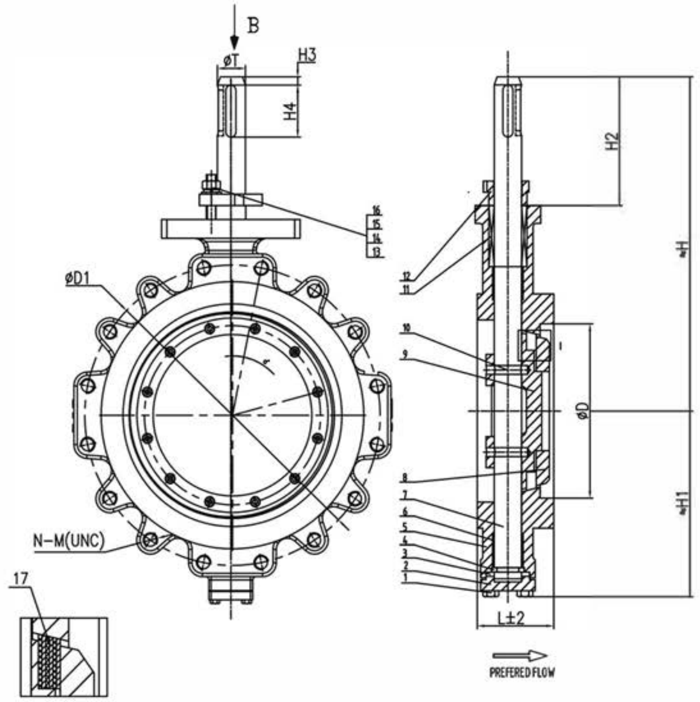
Valve Size	Gear	Inserts
3"	G04.5.6	17x14
4"	G08.10	22x17
6"	G012.14	22x17
8"	G016.18	27x22
10"	G016.18	27x22
12"	G020	36x27

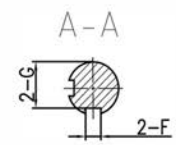
MAX SEAL Valves & Controls offers a broad line of automation systems for precise proportioning or on-off control.

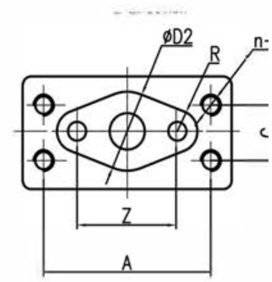
Note: The above actuator mounting dimensions are applicable for wafer, lugged and flanged type valves of pressure rating Pn!O, Pnl6, Pn25, 150Lb, unless otherwise specified.

Butterfly Valve | **Class #300**

Dimensional Data Actuator Mounting With Stem Keyway







Verify Mounting Dimensions Before Manufacturing Mounting Hardware.

Bare Shaft Valvels

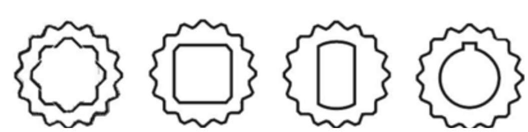
Size mm/inch	L			Actuator Mounting Dimensions																			Torque (in-lbs)	Weight (lbs)
	Wafer	Lug	Flanged	øD	øD1	N-M	a°	H	H1	H2	H3	H4	øT	G	F	A	Z	n-m	C	øD2	R			
3”	64/2.52	48	CF	84	168.3	8”-3/4”-10”	22.5°	251	152	95	5	25	20	16.5	6	90	44	4-M10	40°	32	R9		1770	18
4”	64/2.52	54	CF	105	200	8”-3/4”-10”	22.5°	275	175	100	5	35	25	21	8	90	55	4-M10	40°	40	R11		2655	CF
6”	76/2.99	59	CF	156	270	12”-3/4”-10”	15°	325	220	110	5	40	30	25	10	110	70	4-M12	45°	52	R13		5664	51
8”	89/3.50	73	CF	204	330.2	12”-7/8”-9”	15°	380	268	130	5	50	35	30	10	130	80	4-M16	50°	60	R15		10355	78
10”	114/4.49	83	CF	256	387.4	16”-1”-8”	11.25°	460	308	150	5	50	40	35	12	134	120	4-M16	64°	75	R16		16903	115
12”	144/4.49	93	CF	300	450.8	16”-1 1/8”-8”	11.25°	495	352	150	5	60	45	39.5	14	134	102	4-M16	64°	75	R16		25311	164
14”	127/5.00	92	CF	335	514.5	20”-1 1/8“-8”	9°	555	367	175	5	80	55	49	16	175	120	4-M20	70°	90	R16		31506	217
16”	140/5.51	133	CF	400	571.5	20”-1 1/4“-8”	9°	592	414	180	5	80	60	53	18	175	120	4-M20	70°	90	R16		49605	CF

Gear OPS + 300 Class Valve

Valve Size	Gear	Inserts
3"	G04.5.6	N/A
4"	GOB.10	22x17
6"	G012.14	N/A
8"	G016.18	N/A
10"	G016.18	36x27
12"	G020	36x34

Replaceable Drive Insert Design:

Standard gears are fitted with square drive insert or double D, but a variety of other inserts such as key way are also available. Inserts can be easily replaced or remachined as needed.



Standard Options

For additional information please consult Max-Gear TB 151-20

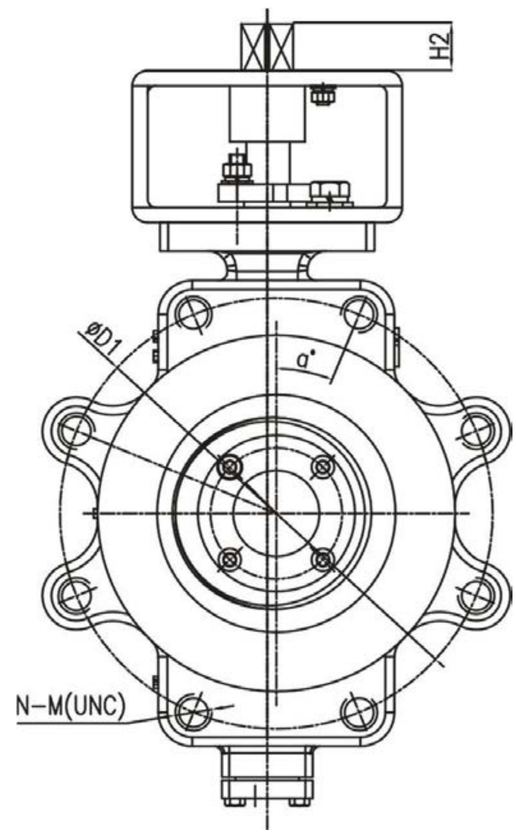
Note: The above actuator mounting dimensions are applicable for wafer, lugged and flanged type valves of pressure rating Pn40, 3001b, unless otherwise specified.

Actuator Mounting Information

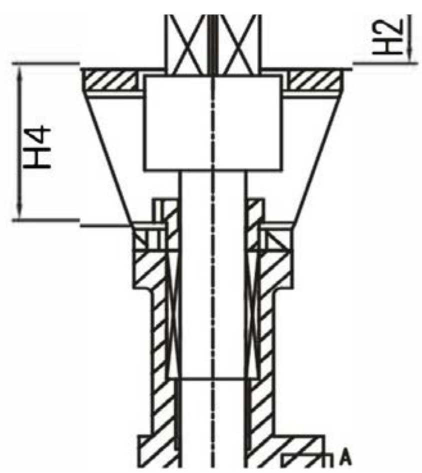
TOV Series Are Suppled With Cast Heavy Duty Actuator Brackets With ISO 5211

Size	ASME 150LB						ASME 300LB					
Inch/mm	Torque NM	Torque In-lb	Top Flanged IOS 5211	H4	H2	Shaft Ends K/d	Torque NM	Torque In-lb	Top Flanged IOS 5211	H4	H2	Shaft Ends K/d
3" / 808	6	761	F07/F10	90	14	14mm x 14mm	200	1770	F07/F10	110	14	14mm x 14mm
4" / 100	160	1416	F07/F10	90	14	14mm x 14mm	300.20	2655	F07/F10	130	17	17mm x 17mm
6" / 150	275	2436	F10/F12	120	17	17mm x 17mm	640	5664	F10/F12	145	22	22mm x 22mm
8" / 200	475	4204	F12/F14	130	22	22mm x 22mm	1170	10355	F12/F14	175	27	27mm x 27mm
10" / 250	795	7035	F12/F14	145	22	22mm x 22mm	1909	16903	F12/F14	175	27	27mm x 27mm
12" / 300	1375	12168	F14/F16	145	27	27mm x 27mm	2860	25311	F14/F16	180	34	34mm x 34mm
14" / 350	2000	17699	F14/F16	170	27	27mm x 27mm	3560	31506	F14/F16	200	36	36mm x 36mm
16" / 400	3000	26549	F16/F25	170	34	34mm x 34mm	5606	49605	F16/F25	200	40	40mm x 40mm
18" / 450	3750	3186	F16/F25	200	40	40mm x 40mm	CF	CF	CF	CF	CF	CF
20" / 500	5100	45133	F25/F30	200	46	46mm x 46mm	CF	CF	CF	CF	CF	CF
24" / 600	6800	60177	F25/F30	225	55	55mm x 55mm	CF	CF	CF	CF	CF	CF

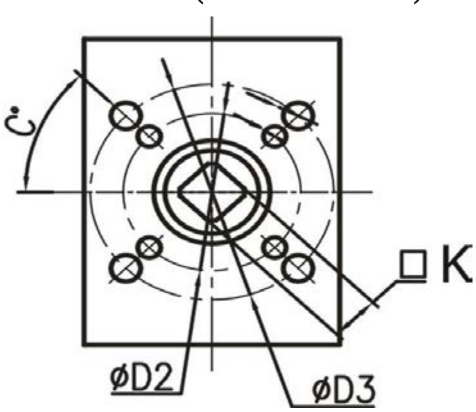
Notes: *Other sizes & pressures are available upon request. *Torque value excluded safety factor for reference. *Up to 20", shaft ends in square type *24"and above, valves' shaft in round type.



Model:
BWT150/BWT300
BLT150/BLT300



Actuator Mounting Pad
B-direction (NPS 3"-NPS 14")



Triple Offset Butterfly Valve

Class 600 and 900 both
Flanged and BUTT Weld Designs
see Z-Tek Series, TB 505-21

Valve Top Work with ISO Mounting
Note: All valves are supplied with ISO 5211 Mounting Bracket & Coupler as standard Wide industrial application range served by Max-Seal product including Chemical/Petrochemical/Oil & Gas/Pulp & Paper/Biochemistry/Pharmaceutical/Water/Mining
Max-Seal reserves the right to change the design and specifications without prior notice. Please consult Max-Seal for details.

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Reivsion No: 503-21_04
Last Review/Update: 03/16/22